



Effect of earthworms in reduction and fate of antibiotic resistant bacteria (ARB) and antibiotic resistant genes (ARGs) during clinical laboratory wastewater treatment by vermifiltration

Sudipti Arora^{a,*}, Sakshi Saraswat^b, Ankur Rajpal^c, Harshita Shringi^a, Rinki Mishra^a, Jasmine Sethi^a, Jayana Rajvanshi^a, Aditi Nag^a, Sonika Saxena^a, A.A. Kazmi^c

^a Dr. B. Lal Institute of Biotechnology, Malviya Industrial Area, Malviya Nagar, Jaipur 302017, India

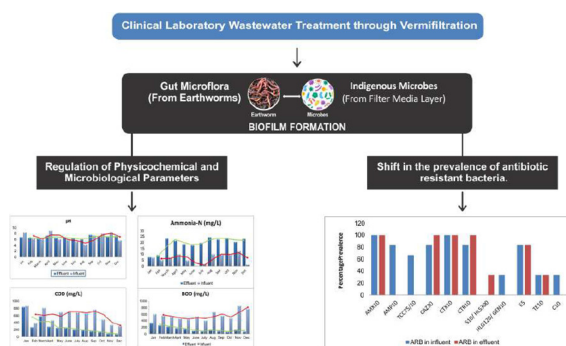
^b Institute of Environmental and Occupational Health Sciences, School of Medicine, National Yang Ming University, Taiwan

^c Environmental Engineering Group, Department of Civil Engineering, Indian Institute of Technology Roorkee, India

HIGHLIGHTS

- Vermifilter is operated for clinical laboratory wastewater treatment for one year.
- VF showed high performance for organics (75–85%) and pathogens removal (99%).
- Earthworm-microorganism interactions help biofilm formation resulting in paradigm shift in ARB.
- Molecular profiling of ARGs was done to understand AMR pattern during vermifiltration.

GRAPHICAL ABSTRACT



ARTICLE INFO

Article history:

Received 29 October 2020

Received in revised form 6 January 2021

Accepted 6 January 2021

Available online 6 February 2021

Editor: Ewa Korzeniewska

Keywords:

Antimicrobial resistance
Antibiotic resistance genes
Clinical laboratory wastewater
Earthworms
Pathogens
Vermifilter
Wastewater treatment

ABSTRACT

In the recent decades, the role of wastewater treatment plants has been entrenched for the dissemination of antibiotic resistant bacteria into the environment. The present study explores the dynamics of earthworms-microorganisms interactions involved in the high treatment efficacy of vermifiltration technology along with reduction of antibiotic resistant bacteria (ARB). This study is the first of its kind to investigate the performance efficacy of vermifilter (VF) for clinical laboratory wastewater treatment. The results of the study showed that earthworms and VF associated microbial community had a significant effect on Biochemical Oxygen Demand (BOD) and Chemical Oxygen Demand (COD) reduction (78–85%), coliforms and pathogen removal (>99.9%) and caused a significant shift in the prevalence pattern of ARB. Molecular profiling of resistance causing genes such as ESBL (*bla_{SHV}*, *bla_{TEM}* and *bla_{CTX-M}*), MRSA (*mec-A*) and Colistin (*mcr-1*) confirmed the probable mechanisms behind the resistance pattern. The microbial community diversity in the influent, earthworm's coelomic fluid and gut and filter media layers associated with the VF assists in the formation of biofilm, which helps in the removal of pathogens from the wastewater. This biofilm formation further results in a paradigm shift in the resistance profile of ARB and ARG, specifically most effective against drugs, targeting cell wall and protein synthesis inhibition such as Ampicillin, Ticarcillin, Gentamicin and Chloramphenicol. These findings further validate vermifiltration technology as a sustainable and natural treatment technology for clinical laboratory wastewater, specifically for the removal of pathogens and antibiotic resistance.

© 2021 Elsevier B.V. All rights reserved.

* Corresponding author at: Dr. B. Lal Institute of Biotechnology, 6E, Malviya Industrial Area, Malviya Nagar, Jaipur 302017, India.
E-mail address: sudiptiarora@gmail.com (S. Arora).

1. Introduction

Over the last few decades, concerns regarding the release of environmental contaminants have rapidly increased. Emerging organic contaminants and xenobiotics such as antibiotics possess a major threat to the environment. The presence of antibiotics is of special concern due to the development of antibiotic resistant bacteria (ARB) and antibiotic resistance genes (ARGs). These substances are extensively used in both human and veterinary medicine against microbial infections and are excreted from the body of the treated organisms, together with their metabolites, within a few days of consumption. Recent studies have shown incomplete metabolism in humans and improper disposal of antibiotics to wastewater treatment plants (WWTPs). It has also been established that these conventional WWTPs form the main source of antibiotic release into the environment as they are insufficient for the exclusion of antibiotics, ARGs and ARBs (Kumar et al., 2019; García et al., 2020). Malpractices such as direct discharge of untreated effluents into the public sewers is being followed by developing nations regardless of the strict regulations issued by concerned authorities such as World Health Organisation (WHO) (Bull World Health Organ, 2001), and Central Pollution Control Board (CPCB) (CPCB, 2001). Since, municipal WWTPs are not designed to deal with these kind of biological waste, and pathogens, these antibiotics are known to survive in the urban WWTPs and, subsequently, in the treated effluent (Fouz et al., 2020). Once in the environment, antibiotics can lead to the continuous selection for ARB that contains ARGs. These microorganisms and pathogens confer resistance against antibiotics by the approach of horizontal gene transfer (Salcedo et al., 2015). Existence of mobile genetic elements such as transposons and integron-associated gene cassettes also contribute towards the spread of ARGs within bacterial communities (Ragan and Beiko, 2009). In urban areas, the regular discharges of WWTP effluents, such as the hospital wastewater effluents, into nearby aquatic bodies like rivers, lakes and oceans prove to be detrimental for the marine life and leads to ecosystem imbalance. Due to the global rise in antibiotic resistance, additional emphasis is now being placed on the detection and removal of pharmaceuticals, ARB and ARGs, and pathogens from treated wastewater but federal and state monitoring programs for these pollutants remains minimal or nonexistent (Beattie et al., 2020). It has been reported that estrogens can cause hermaphroditic fish, while some painkillers are poisonous to trout and certain psycho pharmaceuticals can affect fish and bird behavior (Krarup et al., 2015). Thus, it is highly recommended to perform preliminary onsite treatment for these effluents.

It should be noted that clinical wastewater contains various types of ARGs, including those resistant to antibiotics such as tetracyclines, β -lactams, sulphonamides, macrolides, fluoroquinolones, and even multiple drug resistance (Wang et al., 2020; Laht et al., 2014). Different types of ARGs confer microbial resistance to different antibiotics (Bueno et al., 2018; Lin et al., 2017). In spite of this hazardous influence, there are only few studies investigating the release and direct impact of hospitals and clinical laboratory wastewater into the environment or communal sewage system. Centralized technologies like membrane bioreactor combined with activated sludge process are gaining wide acceptance for hospital and clinical laboratory wastewater treatment due to its high removal capacity for bacteria and antibiotics. However, installation, operation, and maintenance cost form a major challenge towards the establishment of these technologies (Momeni, 2020). Thus, it is imperative to develop and utilize an efficient and cost-effective technology.

Lately, wastewater treatment through earthworm assisted vermifiltration technology, has emerged successful in treating domestic wastewater (Arora et al., 2014; Li et al., 2009), industrial wastewater like herbal pharmaceutical wastewater (Dhadse et al., 2010), gelatine wastewater (Ghatnekar et al., 2010) and dairy wastewater (Sinha et al., 2008) etc. Vermifiltration is a natural, sustainable and emerging green technology which utilizes the potential of earthworms and microorganisms for the treatment of wastewater. It is nature-based, bio-

oxidative process in which earthworms interact intensively with microorganisms, accelerating the stabilization of organic matter and greatly modifying its physical and biochemical properties, resulting in removal of contaminants. The present study investigates the impact and efficiency of earthworms in contaminants removal, pathogens and ARB removal efficiency of an on-site vermifilter for treating clinical laboratory wastewater. The effect of earthworms in treating clinical laboratory wastewater and their impact on ARB and ARG occurrence and further concentrations in influent and effluent are compared to identify potential differences. A clinical diagnostic laboratory deals with urine, blood and faecal samples of infected patients, there are the chances of detection of antibiotics present in their samples, and it is imperative to know the resistance pattern of the microorganisms present in the influent and effluent. In view of this, the present study was carried out to understand the mechanistic insights into vermifiltration technology for clinical wastewater treatment. The main objective of the present study is to determine the performance efficacy of vermifilter and to understand the mechanistic insights by deciphering the role of earthworms in the treatment process. This study also aimed to construe the role of earthworms in AMR and ARB reduction. To the best of our knowledge, this study reports for the first time, the efficacy of the vermifiltration process for the treatment of clinical laboratory wastewater and its effect on antibiotic resistance.

2. Materials and methods

2.1. Experimental design

The study was performed at Dr. B. Lal Institute of Biotechnology, Jaipur, India. A pilot-scale vermifilter (VF) of 1 KLD treatment capacity (material - concrete) with dimensions of 1 m³ was installed at Dr. B. Lal Clinical Laboratory Pvt. Ltd. Campus, Jaipur, in field conditions for treating clinical laboratory wastewater over a period of one year (Jan 2019–Dec 2019). The wastewater was collected by a pump to a collection tank, which was then transferred to a pretreatment or overhead tank, from where the wastewater flowed to the VF by gravity and was then connected to an effluent collection tank. The pretreatment tank contained a mixer for constant mixing of wastewater and sludge and a static ball for controlling water level. The wastewater distribution system in the VF consisted of polyvinyl chloride (PVC) perforated pipes. Fig. 1 describes the working principle of VF along with the design parameters following the methodology described by Arora et al. (2014). The VF bed comprised of different filter media. It was made up of 3 layers of river bed materials, including a layer of coarse size gravel (aggregate size 12–14 mm, 15 cm thickness) as bottom supporting layer, a layer of medium size gravel (aggregate size 6–8 mm, 15 cm thickness) and a layer of fine gravel (aggregate size 2–4 mm, 15 cm thickness) at the top. An aeration space of 15 cm was kept at the top followed by 30 cm layer of fine vermigratings and cow-dung (0.05–5 mm). The active layer containing vermigratings had a uniformity coefficient of 1.36, the effective size of 0.118 mm and a density of 1517.6 kg/m³. It had an average pH of 6.5–7.0, a density of 96 kg/m³ containing lignocellulosic material and the VF rarely produced odour during its long-term operation period, which showed that the VF was aerobic throughout the process. Vermigratings consisted of partially decomposed cow dung. This was prepared and obtained from a separate vermicompost facility unit situated in the campus of Dr. B. Lal Institute of Biotechnology, Jaipur. The VF was inoculated with earthworm's species *Eisenia fetida* with stocking density of 10,000 worms/m³. The VF was supplied the clinical wastewater with a hydraulic loading rate (HLR) of 1.0 m³/m²/day for 12 h continuously and hydraulic retention time (HRT) of 7–8 h. HRT is the time of interaction (residence time/contact time) of wastewater within the filter media layer of vermifilter in which earthworms reside and actual contact time between wastewater & earthworms, whereas; HLR is the rate of application of wastewater to the unit area of filter bed for unit time. HRT is also dependent on the

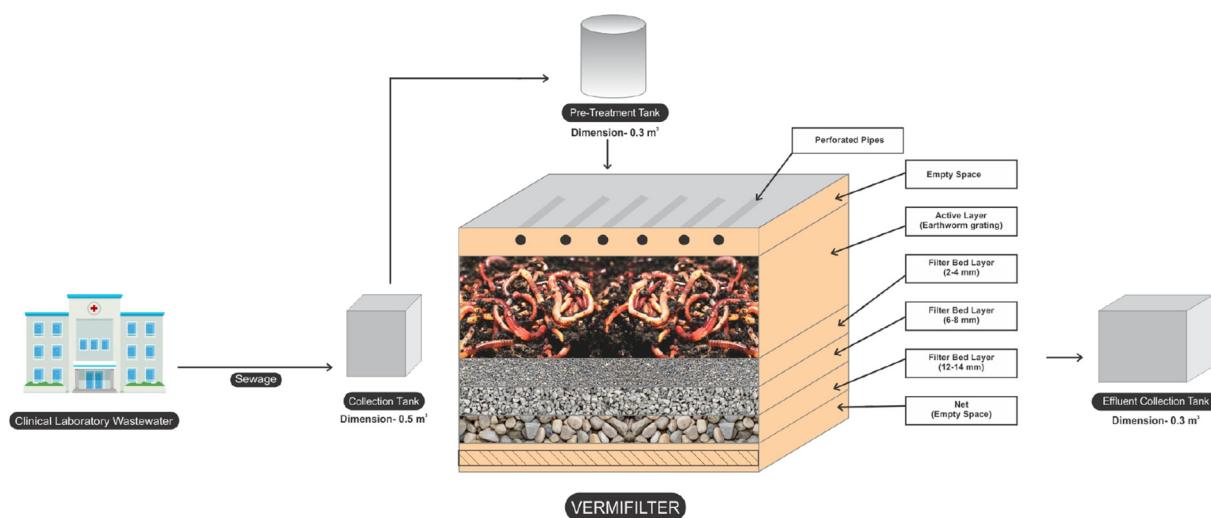


Fig. 1. Schematic flow diagram of vermifilter with descriptions.

filter bed material characteristics. The temperature in the VF bed was monitored under standard conditions, with temperature maintained throughout the study period, ranging between 20 and 30 °C.

2.2. Performance evaluation of vermifilter

The performance of the vermifilter was evaluated by physicochemical and microbiological analysis of influent and effluent samples. Sampling was performed once a week for a period of one year. The physical parameter included pH, temperature, and electrical conductivity (EC). The chemical parameter included Dissolved Oxygen (DO), Total Dissolved Solids (TDS), Salinity, Biochemical oxygen demand (BOD) and chemical oxygen demand (COD), and nutrients (Ammonium (NH_4^+) and Nitrate (NO_3^-)). The microbiological parameter included Total coliforms (TC), Faecal coliforms (FC), Faecal streptococci (FS), *E. coli* and *Salmonella*. All the parameters were analyzed as per the methodology described by Arora et al. (2020). All the experiments were conducted in triplicate and their mean values were recorded. The analysis was performed on the same day of sampling and when the same-day analysis was not possible, samples were stored at 4 °C for less than 24 h in sterile container bottles which were screwed tightly to avoid external contamination. A paired sample *t*-test was performed for each parameter to analyze the differences using SPSS_ statistical package (window version 14.0). The results reported in the study have significance level (P) less than 0.05.

2.3. Investigation of microbial community diversity

Culture based method was used for the detection of microbial community diversity present inside the vermifilter. Isolation of microorganisms from the influent, effluent, earthworm gut, coelomic fluid (CF) and filter media was done by spread plate technique. Sampling was done twice during the operation period. The bacterial colonies were isolated on nutrient agar media and incubated at 30 °C for 24 h. Well-isolated colonies were selected and sub cultured and cells from the new colonies were then picked up with an inoculating needle and transferred onto agar slants to be maintained as pure culture. Pure bacterial cultures were then identified and classified morphologically (shape, gram staining and motility) and biochemically (sugar utilization, indole production, citrate utilization, methyl red-Voges Proskauer (MR-VP), triple sugar iron (TSI) utilization, oxidase production, catalase production, gas production from glucose and coagulase test) according to Bergey's Manual (Bergey's Manual® of Systematic Bacteriology, 2010).

2.4. Determination of antibiotic susceptibility using double disk synergy test

Disk diffusion method was used for antibiotic susceptibility test (Hudzicki, 2012). The isolates obtained from influent and effluent were swabbed onto the surface of Mueller Hinton Agar (MHA) media plate (diameter – 150 mm). The filter paper disks impregnated with a standardized concentration of antibiotics were placed on the surface, and the size of the zone of inhibition around the disk was measured after overnight incubation at 35 °C. The 12 antibiotic disks used are mentioned in Table 1. These are further categorized upon the basis of their class and the mechanism involved in their antibacterial action. These antibiotics were chosen because they are broad spectrum antibiotics used to treat a variety of commonly occurring infections in hospitals and clinical laboratories and, thus, are also routinely discharged in hospital and clinical laboratory wastewater. Each of these antibiotics has its specific mechanism, and they were maintained according to the guidelines outlined in the Clinical and Laboratory Standards Institute [CLSI] documents (Clinical and Laboratory Standards Institute [CLSI], 2017).

2.5. Molecular profiling of ARB genes

To further understand the underlying molecular mechanism, identification of the production of ESBL (extended spectrum beta-lactamases), MRSA (methicillin-resistant *Staphylococcus aureus*) and Colistin genes was performed using culture independent method

Table 1

List of antibiotics used for the detection of ARB along with their class and mechanism of action.

Antibiotics	Class of antibiotics	Mechanism of action
Amoxicillin (AMX ¹⁰) Ampicillin (AMP ¹⁰) Ticarcillin (TCC ^{75/10})	Penicillins	Inhibition of cell wall
Ceftazidime (CAZ ³⁰) Cefotaxime (CTX ¹⁰) Ceftriaxone (CTR ¹⁰)	Cephalosporins	
Streptomycin (S ¹⁰) Gentamicin (HLG ¹²⁰)	Aminoglycosides	Protein synthesis inhibitors
Erythromycin (E ⁵)	Macrolide	
Tetracyclines (TE ¹⁰)	Tetracyclines	Inhibition of nucleic acid
Chloramphenicol (C ¹⁰)	Chloramphenicol	
Ciprofloxacin (CIP ¹⁰)	Fluoroquinolones	

(Drieux et al., 2008). Metagenomic DNA was extracted using the Power water DNA Isolation Kit (Qiagen) following the manufacturer's protocol. This kit is specifically designed to isolate DNA from filtered environmental water samples containing a high level of contaminants. The protocol included an incubation at 65 °C for 10 min following the addition of solution P1 and lysozyme digestion (final concentration = 10 mg/ml) at 37 °C for 1 h. This was followed by RNase digestion at 37 °C for 20 min. Nanodrop-1000 Spectrophotometer (Nanodrop Eppendorf) was used to estimate DNA quantity and quality. DNA concentrations ranged from 75 to 130 ng/μl per sample. The presence of bla_{TEM}, bla_{SHV}, bla_{CTX-M} (ESBL), mec-A (MRSA), and mcr-1 (Colistin) genes in transformants were confirmed by PCR. Approximately 2–3 μg of total DNA for each sample was used for a PCR run. Primers for genotyping were obtained from Eurofins by custom order. The primer pair used for amplification consisted of 27F (5'-AGA GTT TGA TC[A/C] TGG CTC AG-3') and 1492R (5'-G[C/T]T ACC TTG TTA CGA CTT-3').

3. Results and discussions

3.1. Analysis of physicochemical and microbiological parameters for the evaluation of performance efficacy of VF

The performance efficiency was checked thoroughly by analysis of physicochemical and microbiological parameters. For all the physicochemical parameters carried out over a period of 12 months, the average value was calculated and it has been depicted in Figs. 2 and 3. Radical variations were observed in the pH of the influent ranging from 4.2 to 9.3 as clinical wastewater contains a variety of chemical hazardous substances such as pharmaceuticals, radionuclides, pathogens, radioisotopes, solvents, endocrine disruptors, detergents, and disinfectants. Despite this, it was gradually neutralized to the range of 6.7–7.0

in the effluent, referring to the avid buffering capacity of the earthworms (Fig. 2a) (Sinha et al., 2008). There was a significant variation (t -test: $P < 0.05$) in the pH of the effluent samples after achieving the steady state. Temperature is another key factor that affects earthworm but it was found to be in the range of 20–30 °C (Fig. 2b) even during extreme summers and winters and didn't show any harsh effect on the earthworms. This consistency was mainly because of the material used for the construction of VF tank and river bed materials as filter media. The average EC in the influent and effluent seemed to lie in the range of 2.9–3.1 mS/cm (Fig. 2c). No significant difference was observed between the average salinity of the influent and effluent throughout the study (t -test: $P < 0.05$) as shown in Fig. 2d. TDS concentration in VF effluent was reduced by 30% from the influent after completion of the treatment process (Fig. 3a). Earthworms are reported to ingest organic and inorganic solid particles in wastewater and excrete them as finer particles. These finer particles are further trapped in the voids of VF and later are degraded by microbial actions. Due to this reason, VF does not choke throughout the operation period but rather works smoothly and uninterruptedly, which makes it an excellent choice (Arora et al., 2016). DO is another important parameter which had the mean concentration of 0.53 mg/l in the VF influent. In VF effluent, the mean DO was observed to be 3.1 mg/l (Fig. 3b). DO signifies the environmental conditions prevailing inside the VF and it is one of the significant parameters in outflow because low oxygen level is toxic to earthworms and prevailing aerobic microorganisms. A significant variation was observed in DO of effluent samples during the study period (t -test: $P < 0.05$). The increase in DO is also attributed to the dropping overflow of the VF layers, as water flowed through it naturally by gravity and earthworm also provide aeration through their burrowing activity in the filter bed (Wang et al., 2011). Higher DO in the effluent reduces the septic condition and brings a good chance of wastewater reuse for

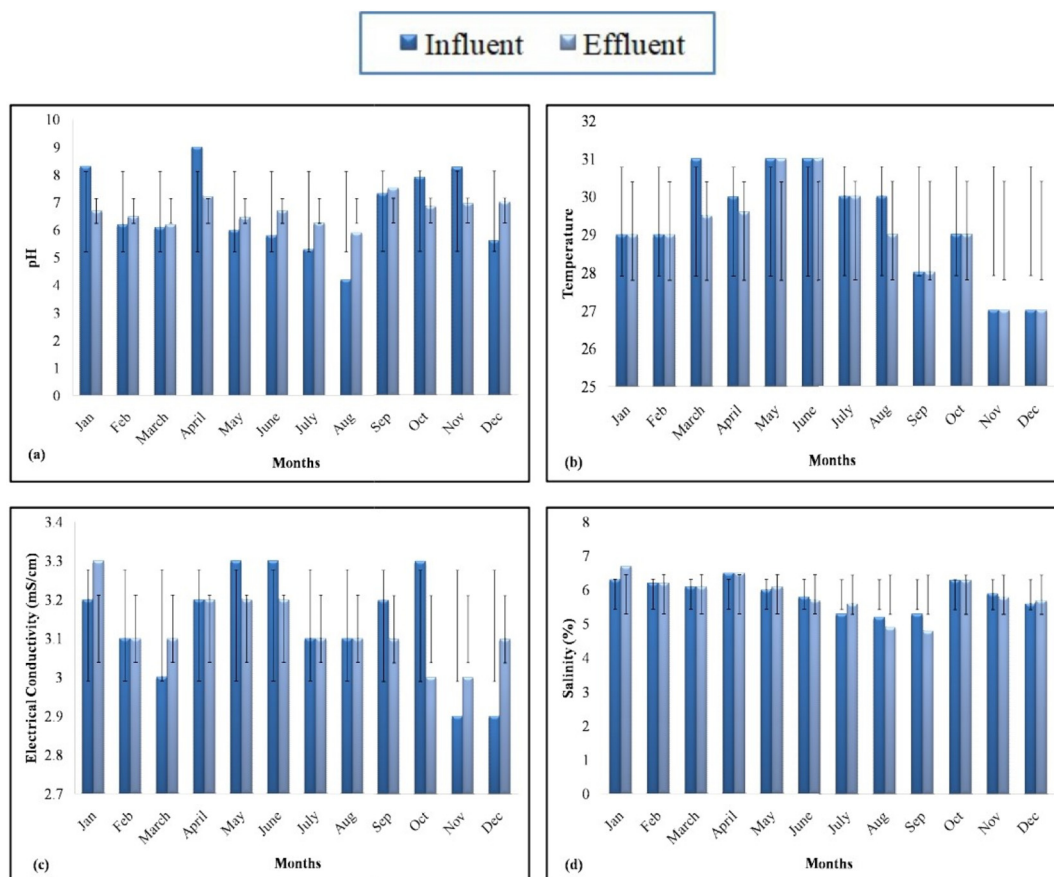


Fig. 2. Variations in (a) pH, (b) temperature, (c) EC and (d) salinity over a period of one year. All values represent average of triplicates (mean $\pm$ standard deviation, $n = 3$).

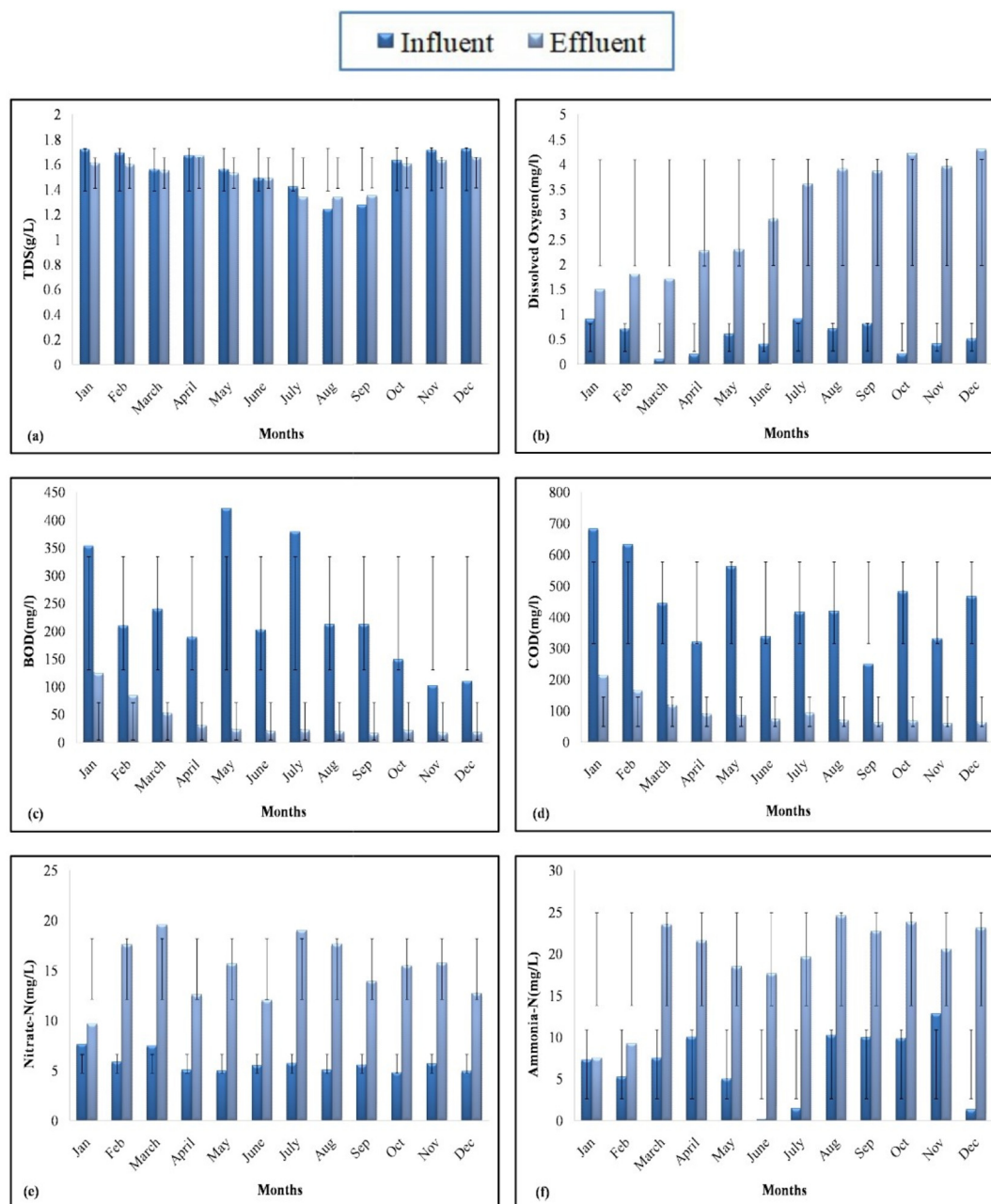


Fig. 3. Variations in (a) TDS (b) DO (c) BOD (d) COD (e) Nitrate-N (f) Ammonia-N over a period of one year. All values represent an average of triplicates (mean $\pm$ standard deviation, $n = 3$).

irrigation purposes (Singh et al., 2019). DO content is also closely related to organic matter removal calculated in terms of BOD and COD. These are two most essential parameters which define the profile of any type of water. The average BOD in the influent that was observed to be in the range of 200–250 mg/L, which was reduced to less than 30 mg/L in the effluent after the acclimatization period (Fig. 3c) whereas the COD was found to be around 390–420 mg/l in the influent decreased to 60–90 mg/l in the effluent (Fig. 3d). In VF, the COD and BOD concentration were reduced significantly (t -test: $P < 0.05$). These results show decrement by 80–85% and 75–80% for BOD and COD, respectively. The reason for high BOD and COD removal is the reduction in organic load present in the influent, by the enzymes produced by the earthworms and microorganisms such as cellulose, amylase and protease. Earthworms also have the ability to enhance the biodegradation of organic matter through their feeding, burrowing and casting behavior. They host millions of decomposer microbes in their gut and excrete them

along with nutrients like nitrogen in their excreta. Nitrogen is one of the predominant contaminants in sewage, and nitrates and ammonia result in water pollution and eutrophication (Liu et al., 2012). The average concentration of NO_3^- -N in influent was 5.67 mg/l. This concentration experienced an increment of 62% in the effluent (Fig. 3e). Similarly, NH_4^+ -N in the influent was detected in the range of 6.6–7.0 mg/l, while effluent increased to 18–19 mg/l (Fig. 3f), which is significantly high (t -test: $P < 0.05$). This could be attributed to the rapid growth of ammonia oxidizing bacteria at a temperature above 15 °C (Liu and Wang, 2013). NH_4^+ -N was subsequently converted to NO_3^- -N via biological nitrification, carried out by aerobic autotrophic bacteria using molecular oxygen as an electron acceptor (Wang et al., 2011). Additionally high surface DO concentration also proved to be beneficial for the survival of aerobic microbes, which was advantageous for nitrification. All these parameters collectively result in a higher treatment efficiency of the VF.

Table 2
Changes in microbiological parameters of clinical wastewater during vermifiltration.

Parameters	Influent ^a	Effluent ^a	Average percentage removal	Significance level (t-test)
TC (MPN/100 mL)	6.7 ± 0.7	5.5 ± 0.9	90%	P < 0.05
FC (MPN/100 mL)	4.5 ± 0.5	2.5 ± 0.5	99%	P < 0.05
FS (MPN/100 mL)	4.3 ± 0.3	2.80 ± 0.1	99%	P < 0.05
<i>E. coli</i> (CFU/ml)	3.5 ± 0.5	1.2 ± 0.3	99%	P < 0.05
<i>Salmonella</i> (CFU/ml)	3.8 ± 0.5	1.67 ± 0.92	99%	P < 0.05

^a Arithmetic averages (mean ± standard deviation, n = 3).

The presence of coliforms is often used as an indicator of the overall pathogenicity of the sample. Table 2 illustrates the coliform and pathogen removal efficacy for wastewater after Vermifiltration. It was observed that the concentration of pathogens in VF was reduced considerably. Log Removal Values were determined by taking the logarithm of the ratio of pathogen concentration in the influent and effluent and by utilizing these values, percentage removal was obtained. The average values of TC, FC and FS were reduced by 90%, 99%, and 99% in the effluent, respectively, showing a significant reduction in pathogen (*t*-test: *P* < 0.05) and quality enhancement of the effluent. The log removal of *E. coli* and *Salmonella* in VF effluent was observed to be 3, with the percentage removal of 99.9%. High removal values may be attributed to the release of coelomic fluid from the body cavity of earthworms reported by Arora et al. (2014), which shows antibacterial properties that inhibit the growth of pathogens. Vermifiltration has been shown to reduce pathogens to safe levels through the action of earthworm-microorganisms interactions and antibacterial activity of microflora associated with earthworms (Arora et al., 2016).

3.2. Microbial community diversity associated with earthworm-microorganisms interactions

The results revealed the microbial community diversity, based on culture assay showed the presence of sixty isolates from influent, effluent, earthworm coelomic fluid, gut of earthworms and filter media layer samples as shown in Fig. 4. Out of 60 isolates, 22% belong to the *Micrococcus* sp. (13 isolates), 17% belong to *Bacillus* sp. (10 isolates), 11% belong to *Corynebacterium* sp. (7 isolates), 10% belong to *Proteus* sp., *Pseudomonas* sp. and *Staphylococcus* sp. (6 isolates each), 5% belong to *Shigella* and *E. coli* each (3 isolates each), 3% belong to *Citrobacter* sp. and *Klebsiella* sp. (2 isolates each), and 2% belong to *Enterobacter* and *Providencia* (1 isolate each). All these microorganisms occur in a wide range of aerobic to facultative anaerobic bacteria. These results are analogous to the previous studies and contribute to efficient working of a VF system (Arora et al., 2016; Wang et al., 2011). Clinical wastewater outlines the source of various pathogenic and opportunistic pathogens in the influent like *Proteus*, *Shigella*, *Pseudomonas*, *Bacillus* sp. and *Klebsiella* spp. are amongst the most common causes of a variety of community-acquired and hospital-acquired infections (Lederman and Crum, 2005). *E. coli* strains are part of the normal microbiota of the gut and are expelled into the environment within faecal matter. *Corynebacterium* sp. acts as pathogens in immunosuppressed patients. It was observed that filter media layer and coelomic fluid were primarily inhabited by gram-positive microflora. However, contrary to this, the isolates obtained from the earthworm gut belonged to the gram-negative microflora. Earthworms are reported to stimulate and accelerate microbial activity by increasing the population of microorganisms and through improved aeration by burrowing actions (Khyade, 2018). The selected isolates obtained from influent and effluent samples were further subjected to antibiotic susceptibility testing.

3.3. Antibiotic susceptibility testing

The study identified culturable aerobic microbial community composition of influent and effluent that were resistant to antibiotics of

critical importance. A total of 12 strains isolated from influent and effluent were selected to determine antibiotic susceptibility test against the drugs as shown in Fig. 5. The results indicated that none of the species showed resistance against Ciprofloxacin (CIP10/5) in both influent and effluent. Few of the isolates showed equal resistance in the influent and effluent samples against Amoxicillin (AMX10), Cefotaxime (CTX10), Erythromycin (E5) and Tetracyclines (TE10). Interestingly, few species which were found to be resistant in the influent against Ampicillin (AMP10), Ticarcillin (TCC75/10), Gentamicin (HLG120/GEN10) and Chloramphenicol (C10) experienced a decrement in the percentage of resistant isolates in the effluent samples. Another intriguing observation was the presence of an isolate that was resistant against Streptomycin (S10/HLS300) in the effluent, while no resistant isolates were observed in the influent. The probable reason for this can either be residual contaminants during the collection or the presence of non-culturable bacteria. Thus, to determine the source of resistance and to elucidate the mechanism associated with it, further profiling of ARG was conducted for the effluent sample.

3.4. Detection of ARGs for ESBL, MRSA and Colistin resistant strains through molecular characterization

Molecular characterization of different antimicrobial resistance bacteria was done by the presence or absence of various molecular markers like bla_{SHV}, bla_{TEM} and bla_{CTX-M} genes for ESBL, *mecA* gene for MRSA and *mcr-1* gene for Colistin in the effluent samples from the VF. The sample was screened for the presence of such marker genes by PCR using the primer against these genes markers. The results show that only bla_{CTX-M} gene with a band size of 544 bp was amplified, indicating the presence of resistance against ESBL drug group Cephalosporins. However, all other resistant gene markers were not present in the effluent sample. MRSA isolates, having *mecA* gene are reported to be resistant to penicillin and oxytetracycline, whereas no resistance was observed against ciprofloxacin, chloramphenicol and gentamicin (Kumar et al., 2017). This corroborated the antibiotic susceptibility test and even further validated the decrease in resistance in the non culturable species in the effluent samples. The production of ESBLs is one of the most important mechanisms of antimicrobial resistance. Emergence of bla_{CTX-M} producing Enterobacteriaceae is attributed to increase in the number of ESBL production. It is reported that *E. coli* is the predominant CTX-M group I ESBL producing Enterobacteriaceae and resistance to third generation Cephalosporins (3GC) antibiotics is the possible reason for the emergence and spread of CTX-M ESBL producing organisms (Priyadharsini et al., 2011). This result correlates with the antibiotic susceptibility test as high resistance was shown against 3GCs.

S. epidermidis was found in the effluent, which is believed to acts as a reservoir for *mecA* gene and transfer it to *S. aureus* by horizontal gene transfer (Najar-Peerayeh et al., 2014). There is a possibility that this species present in the effluent was sensitive (MSSE) (Bhimraj, 2018) since no bands were observed for *mecA* gene. However, it might also be possible that earthworm-microorganism interactions play a certain role in shifting the resistivity pattern. The *mcr* genes are responsible for horizontal transfer of colistin resistance. This gene is generally present in *Proteus* strains like *P. rettgeri*, *P. vulgaris* and *P. mirabilis* (Aghapour



Fig. 4. Microbial community diversity isolated from influent, effluent, earthworm gut, coelomic fluid and filter media layer.

et al., 2019). However, since *P. inconstans* was isolated from the VF effluent, *mcr* gene was likely to be absent.

3.5. Mechanism responsible for the paradigm shift in antibiotic resistance pattern observed due to earthworm-microorganisms interactions

Vermifiltration is proven to be a natural and sustainable technology for the treatment of wastewater with combined action of earthworms and microorganisms. This study focused on the performance efficacy of vermifilter situated at Dr. B. Lal Institute of Biotechnology for clinical laboratory wastewater. The main objective was to investigate the effect of earthworms on the resistant microbial community found in the clinical wastewater before and after the treatment. It was observed that there is a decrease in overall percentage of drug resistant strains towards the different drugs observed in the VF treated effluent as compared to the influent. This can be explained by the combined action of earthworms and indigenous microbes present in the active layer of the VF. Various microbial communities are responsible for the high removal percentages in the VF effluent (Ghobadi et al., 2016).

Additionally, earthworms represent the dominant invertebrate in soil and promote the decomposition of organic pollutants. They are one of the typical saprozoic organisms, living in the environment replete with microorganisms, some of which may be a threat to their existence. It has been reported that to survive in such an environment, they have developed efficient immune-defense mechanisms against invading microorganisms (Riera Romo et al., 2016). Antibacterial peptides present in earthworms are listed as an important defense component against antimicrobial resistance. Several anti-infection components such as lysozymes and fetidins have been found in the earthworms (Wang et al., 2003). The presence of earthworms can also stimulate microbial activities and population, which in turn, accelerate the degradation of pollutants present in wastewater (Wang et al., 2017). As a result, earthworms and indigenous microbes together form a biofilm and upon exposure to this biofilm, the ARB presents in the influent experiences a shift in the antibiotic resistivity pattern. Biofilm process plays a significant role in breaking down and removing adsorbed organic substances. VF biofilms are exposed to the activities of earthworms, and these conditions influence the microbial metabolism and the transformation of

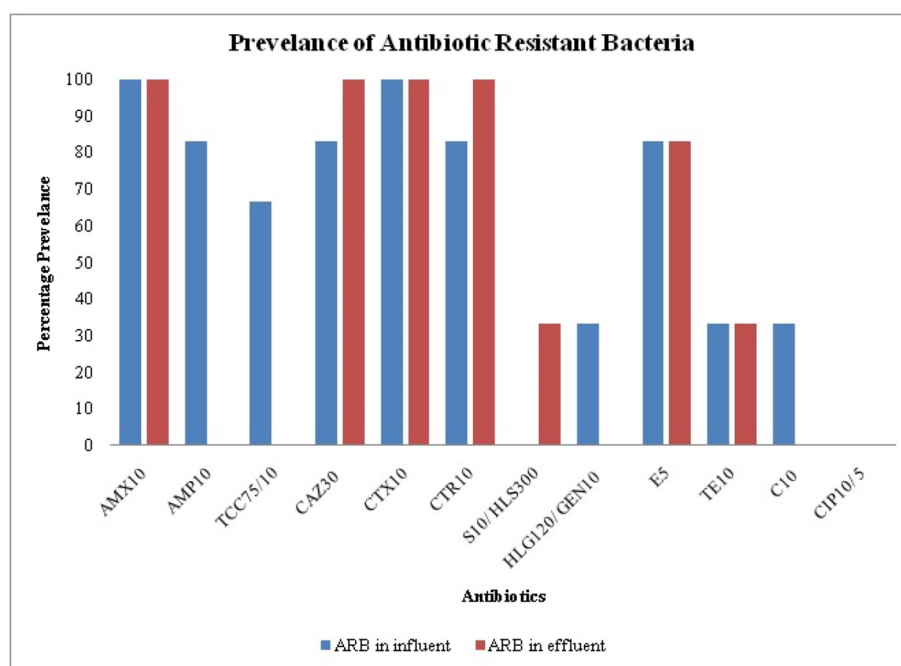


Fig. 5. Pervasiveness of antibiotic resistant bacteria in influent and effluent.

organic matter in the biofilm (Yang et al., 2013). It has also been reported that earthworms could effectively decline tetracycline and fluoroquinolone resistance genes in activated sludge system as compared to the stabilization sludge system and without using earthworms (Huang et al., 2018). The bacteria present in the earthworm gut promote antibiotic degradation. Whereas, in the soil, degradation is mainly carried out by *Pseudomonas*, *Staphylococcus*, and *Planctomycetes* (Sun et al., 2020). Therefore, the abundance of these bacteria strains was significantly increased in the worm gut or the adjacent soil when exposed to antibiotics and antibiotic resistant bacteria. Thus, this shift in the paradigm of wastewater might be due to long term exposure of ARB to the

biofilm formed as a result of the interaction between earthworms and indigenous microbes as illustrated in Fig. 6. However, further research is needed to clarify the degradation mechanism of earthworm's intestinal bacteria on ARB.

Vermifilter proved to be an excellent artificial system through the inoculation of earthworms into a conventional biofilter, which works on the principle of earthworm-microorganism symbiotic and synergistic interactions to treat clinical laboratory wastewater. Earthworm bioturbation impacts on the fate of contaminants by modifying their mobility and sorption to soil, leading to the change of the quality and dynamic of soil organic materials. Their gut also acts as a hotspot for

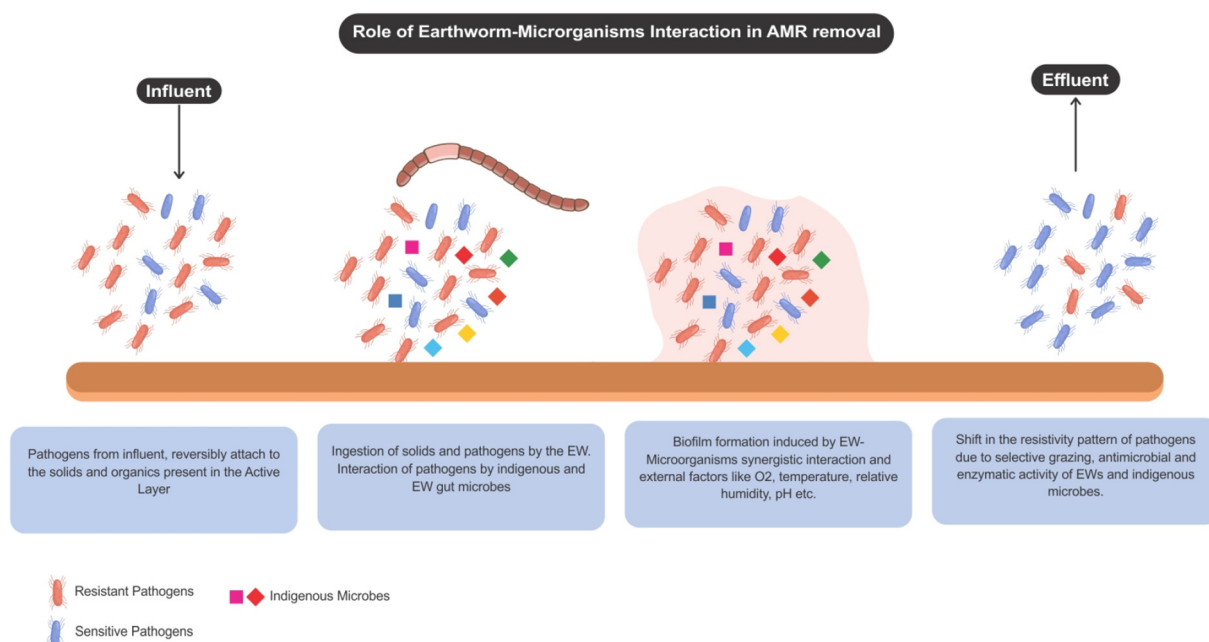


Fig. 6. Earthworm-microorganisms interactions in biofilm formation for reduction of AMR present in clinical laboratory wastewater.

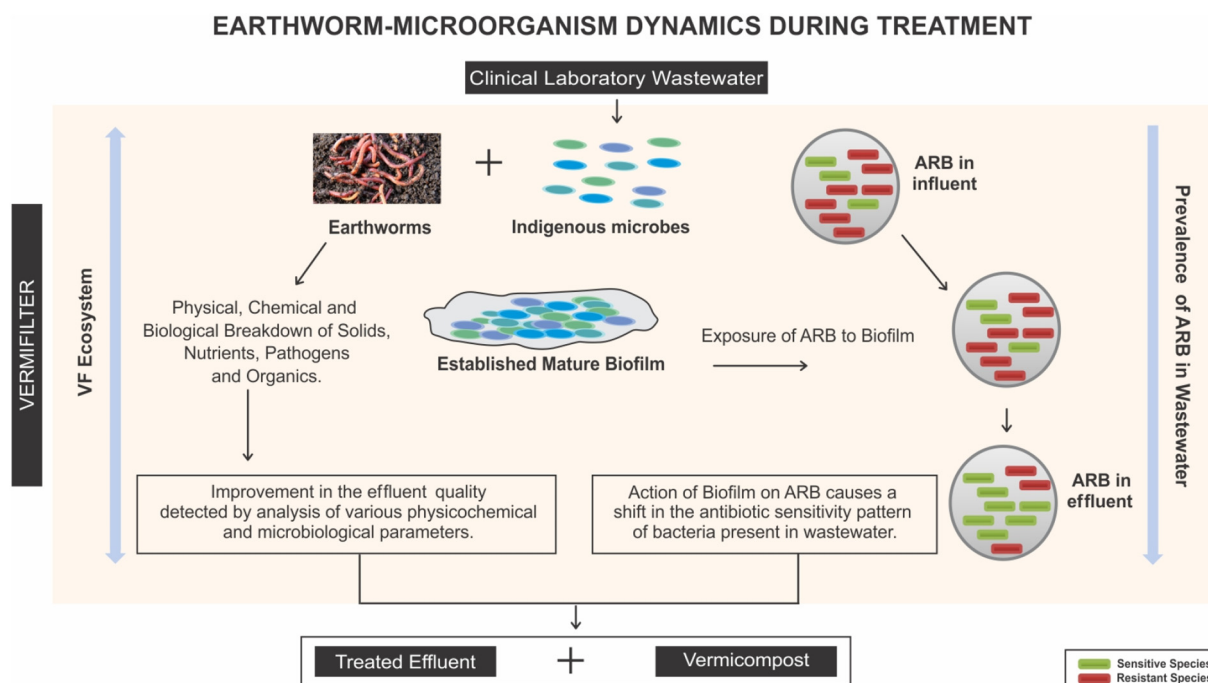


Fig. 7. Dynamics involved in the treatment of clinical laboratory wastewater during the process of vermifiltration.

microbial activity. Additionally, they perform effective degradation and transformation directly, through generating enzymes and, indirectly, owing to the microflora present in their intestine (Ghobadi et al., 2016; Mougin et al., 2013). This is well described in Fig. 7. The fate of antibiotic resistance genes during six different full-scale municipal wastewater treatment processes and revealed that horizontal and vertical gene transfer mainly occurs in aerated tanks, whereas anaerobic or anoxic tanks can reduce the occurrence of antibiotic resistance (Tong et al., 2019). However, a deeper analysis is required to evaluate whether vertical and horizontal gene transfer is the key process for transmission of antibiotic resistance within the VF. An intriguing observation was the improvement in the overall activity of cell wall targeting and protein synthesis inhibiting drugs. Several studies have reported the presence of selective foraging in earthworms for solids, organic particles (Bhadauria and Saxena, 2010), microorganisms and other pathogens (Sinha et al., 2008; Kale and Karmegam, 2010), however, further analysis might clearly reveal their specificity towards ARB.

It should also be noted that the presence of antibiotic compounds is not the only factor influencing the fate of antibiotic resistance. Therefore, other factors should also be taken into accounts such as the operational parameters of VF, the wastewater-associated bacterial communities, and their interaction with different microbial communities, which may contribute to the spread of ARGs associated with microorganisms. Conclusively, VF technology offers a sustainable solution for clinical wastewater with reliable performance in terms of reduction of physicochemical and microbiological parameters for clinical wastewater treatment. Also, since the effluents of hospitals contain high quantities of various ARBs, vermifiltration could be considered as a promising and economical technology for tertiary or refining treatment of these kinds of wastewater (Tehrani and Gilbride, 2018).

It is noteworthy that this study focused on the ARB present in the clinical laboratory wastewater fraction and further studies alluding to antibiotics and antibiotic resistance genes might provide a consolidated result helping in understanding the relationship between earthworm-microorganism activity and ARB. Despite this confinement, the combined culture-based and molecular approach used in this study provides a deeper understanding of the antimicrobial resistance community present in clinical laboratory wastewater.

4. Conclusions

On-site treatment of clinical laboratory wastewater is vital for the health and safety of surface waters used for recreation, drinking water, and animal habitat. The results confirmed the potential of vermifiltration technology for the treatment of clinical laboratory wastewater. Also, the study focused on selecting the potential bacterial strains from influent and effluent that possess antimicrobial resistance. It was found that earthworm-microorganism interactions play a crucial role in the regulation of physicochemical and microbiological parameters and additionally was responsible for causing a shift in the antimicrobial resistance pattern by the help of biofilm formation. The most efficient activity of antimicrobial resistance was found against drugs targeting cell wall and inhibiting protein synthesis. Together, these results suggest that vermifilters prove to be natural and sustainable on-site wastewater treatment technology for clinical laboratory wastewater thus, reducing the dissemination and proliferation of pathogens and ARB entering into the water streams.

Funding information

The present research study is partially funded by the Department of Science & Technology-Gol (Grant No. DST/WTI/2K16/193) for financial support to this research and by internal mural grants (IMG) from Institutional Research Scientific Committee of the institute Dr. B. Lal Institute of Biotechnology, Jaipur.

CRediT authorship contribution statement

Sudipti Arora (Corresponding author): Conceptualization, investigation, resources, manuscript Writing and editing, **corresponding author.**

Sakshi Saraswat: Experimental design, protocol standardization, Experimental conduction, Data analysis, Writing -original draft of manuscript.

Ankur Rajpal: Manuscript review, Statistical analysis.

Harshita Shringi: Sampling, Experimentation, Data analysis.

Rinki Mishra: Experimental design, protocol standardization, Experimental conduction, Data analysis.

Jasmine Sethi: Sampling, Experimentation, Data analysis.

Jayana Rajvanshi: Sampling, Experimentation, Data analysis.

Aditi Nag: Experimental design on molecular work, protocol standardization, Experimental conduction.

Sonika Saxena: Resources providing, Sample collection protocol standardization and supervision.

A. A. Kazmi: Conceptualization and manuscript review.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The study group would like to acknowledge the constant support received from Dr. B. Lal Gupta (Director, Dr. B. Lal Institute of Biotechnology, Jaipur) and Dr. Aparna Datta for inspiring this research and providing daily motivations to work faster. The team further acknowledges the efforts made by research students Ms. Vikky Sinha, Ms. Vaishali Saboo and Mr. Harshit Chhabra for conducting experiments.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.scitotenv.2021.145152>.

References

- Aghapour, Z., Gholizadeh, P., Ganbarov, K., Bialvaei, A.Z., Mahmood, S.S., Tanomand, A., et al., 2019. Molecular mechanisms related to colistin resistance in Enterobacteriaceae. *Infect Drug Resist* 12, 965–975. <https://doi.org/10.2147/IDR.S199844>.
- Arora, S., Rajpal, A., Bhargava, R., Pruthi, V., Bhatia, A., Kazmi, A.A., 2014. Antibacterial and enzymatic activity of microbial community during wastewater treatment by pilot scale vermifiltration system. *Bioresour. Technol.* <https://doi.org/10.1016/j.biortech.2014.05.041>.
- Arora S, Rajpal A, Kazmi AA. Antimicrobial Activity of Bacterial Community for Removal of Pathogens during Vermifiltration. *J Environ Eng (United States)* 2016. doi:[https://doi.org/10.1061/\(ASCE\)EE.1943-7870.0001080](https://doi.org/10.1061/(ASCE)EE.1943-7870.0001080).
- Arora, S., Saraswat, S., Mishra, R., Rajvanshi, J., Sethi, J., Nag, A., et al., 2020. Design. Performance Evaluation and Investigation of the Dynamic Mechanisms of Earthworm-Microorganisms Interactions for Wastewater Treatment Through Vermifiltration Technology 21, 1–9. <https://doi.org/10.1101/2020.08.14.252072>.
- Beattie, R.E., Skwor, T., Hristova, K.R., 2020. Survivor microbial populations in post-chlorinated wastewater are strongly associated with untreated hospital sewage and include ceftazidime and meropenem resistant populations. *Sci. Total Environ.* <https://doi.org/10.1016/j.scitotenv.2020.140186>.
- Bergey's Manual® of Systematic Bacteriology <https://doi.org/10.1007/978-0-387-68572-4>.
- Bhaddauria, T., Saxena, K.G., 2010. Role of earthworms in soil fertility maintenance through the production of biogenic structures. *Appl Environ Soil Sci* <https://doi.org/10.1155/2010/816073>.
- Bhimraj, A., 2018. Postoperative Neurosurgical Infections. Elsevier Inc. <https://doi.org/10.1016/B978-0-323-32106-8.00050-9>.
- Bueno, I., Williams-Nguyen, J., Hwang, H., Sargeant, J.M., Nault, A.J., Singer, R.S., 2018. Systematic review: impact of point sources on antibiotic-resistant bacteria in the natural environment. *Zoonoses Public Health* <https://doi.org/10.1111/zph.12426>.
- Safe management of wastes from health care activities. *Bull World Health Organ* <https://doi.org/10.1590/S0042-96862001000200013>.
- Clinical and Laboratory Standards Institute (CLSI). Performance Standards for Antimicrobial Susceptibility Testing; 27th edition. CLSI supplement M100. 2017.
- CPCB, 2001. *Environmental Standards for Ambient Air. Automobiles, Fuels, Industries and Noise*.
- Dhadse, S., Satyanarayan, S., Chaudhari, P.R., Wate, S.R., 2010. Vermifilters: a tool for aerobic biological treatment of herbal pharmaceutical wastewater. *Water Sci. Technol.* <https://doi.org/10.2166/wst.2010.523>.
- Drieux, L., Brossier, F., Sougakoff, W., Jarlier, V., 2008. Phenotypic detection of extended-spectrum β -lactamase production in Enterobacteriaceae: review and bench guide. *Clin. Microbiol. Infect.* <https://doi.org/10.1111/j.1469-0691.2007.01846.x>.
- Fouz, N., Pangesti, K.N.A., Yasir, M., Al-Malki, A.L., Azhar, E.I., Hill-Cawthorne, G.A., et al., 2020. The contribution of wastewater to the transmission of antimicrobial resistance in the environment: implications of mass gathering settings. *Trop Med Infect Dis* 5. <https://doi.org/10.3390/tropicalmed5010033>.
- García, J., García-Galán, M.J., Day, J.W., Boopathy, R., White, J.R., Wallace, S., et al., 2020. A review of emerging organic contaminants (EOCs), antibiotic resistant bacteria (ARB), and antibiotic resistance genes (ARGs) in the environment: increasing removal with wetlands and reducing environmental impacts. *Bioresour. Technol.* <https://doi.org/10.1016/j.biortech.2020.123228>.
- Ghatnekar et al. 2010. Application of Vermi-filter-based Effluent Treatment Plant (Pilot scale) for Biomanagement of Liquid Effluents from the Gelatine Industry. *Dyn Soil, Dyn Plant* 2010.
- Ghobadi, N., Shokoohi, R., Rahmani, A.R., Samadi, M.T., Godini, K., Samarghandi, M.R., 2016. Performance of a pilot-scale vermifilter for the treatment of a real hospital wastewater. *Avicenna J Environ Heal Eng* <https://doi.org/10.5812/ajehe-7585>.
- Huang, K., Xia, H., Wu, Y., Chen, J., Cui, G., Li, F., et al., 2018. Effects of earthworms on the fate of tetracycline and fluoroquinolone resistance genes of sewage sludge during vermicomposting. *Bioresour. Technol.* <https://doi.org/10.1016/j.biortech.2018.03.021>.
- Hudzicki, J., 2012. Kirby-Bauer disk diffusion susceptibility test protocol author information. *Am Soc Microbiol* 1–13.
- Kale, R.D., Karmegam, N., 2010. The role of earthworms in tropics with emphasis on Indian ecosystems. *Appl Environ Soil Sci* <https://doi.org/10.1155/2010/414356>.
- Khyade VB. Bacterial diversity in the alimentary canal of earthworms. *J Bacteriol Mycol Open Access* 2018. doi:[10.15406/jbmoa.2018.06.00200](https://doi.org/10.15406/jbmoa.2018.06.00200).
- Krurup J, Jakob S, Nielsen U, Stefan Z. New standard for hospital wastewater treatment. *Filtr + Sep* 2015. doi:[https://doi.org/10.1016/s0015-1882\(15\)30141-5](https://doi.org/10.1016/s0015-1882(15)30141-5).
- Kumar A, Kaushik P, Anjay, Kumar P, Kumar M. Prevalence of methicillin-resistant *Staphylococcus aureus* skin and nasal carriage isolates from bovines and its antibiogram. *Vet World* 2017. doi:[10.14202/vetworld.2017.593-597](https://doi.org/10.14202/vetworld.2017.593-597).
- Kumar, M., Jaiswal, S., Sodhi, K.K., Shree, P., Singh, D.K., Agrawal, P.K., et al., 2019. Antibiotics bioremediation: perspectives on its ecotoxicity and resistance. *Environ. Int.* <https://doi.org/10.1016/j.envint.2018.12.065>.
- Laht, M., Karkman, A., Voolaid, V., Ritz, C., Tenson, T., Virta, M., et al., 2014. Abundances of tetracycline, sulphonamide and beta-lactam antibiotic resistance genes in conventional wastewater treatment plants (WWTPs) with different waste load. *PLoS One* 9, 1–8. <https://doi.org/10.1371/journal.pone.0103705>.
- Lederman, E.R., Crum, N.F., 2005. Pyogenic liver abscess with a focus on *Klebsiella pneumoniae* as a primary pathogen: an emerging disease with unique clinical characteristics. *Am. J. Gastroenterol.* <https://doi.org/10.1111/j.1572-0241.2005.40310.x>.
- Li, X., Yang, J., Zhao, L., Xing, M., Deng, D., Yi, D., 2009. Profiling of Microbial Community Diversity in Vermifiltration of Different Earthworm Abundances by PCR-DGGE 25–27.
- Lin, H., Zhang, J., Chen, H., Wang, J., Sun, W., Zhang, X., et al., 2017. Effect of temperature on sulfonamide antibiotics degradation, and on antibiotic resistance determinants and hosts in animal manures. *Sci. Total Environ.* <https://doi.org/10.1016/j.scitotenv.2017.07.057>.
- Liu, G., Wang, J., 2013. Long-term low DO enriches and shifts nitrifier community in activated sludge. *Environ Sci Technol* <https://doi.org/10.1021/es304647y>.
- Liu, J., Lu, Z., Yang, J., Xing, M., Yu, F., Guo, M., 2012. Effect of earthworms on the performance and microbial communities of excess sludge treatment process in vermifilter. *Bioresour. Technol.* <https://doi.org/10.1016/j.biortech.2012.04.096>.
- Momeni M. Hospital wastewater treatment scenario around the globe 2020; 21:1–9.
- Mougin, C., Cheviron, N., Repincay, C., Hedde, M., Hernandez-Raquet, G., 2013. Earthworms highly increase ciprofloxacin mineralization in soils. *Environ. Chem. Lett.* <https://doi.org/10.1007/s10311-012-0385-z>.
- Najar-Peerayeh, S., Moghadas, A.J., Behmanesh, M., 2014. Antibiotic susceptibility and mecA frequency in *Staphylococcus epidermidis*, isolated from intensive care unit patients. *Jundishapur J Microbiol* <https://doi.org/10.5812/jjm.11188>.
- Priyadharsini, R.I., Kavitha, A., Rajan, R., Mathavi, S., Rajesh, K.R., 2011. Prevalence of bla CTX M extended spectrum beta lactamase gene in Enterobacteriaceae from critical care patients. *J Lab Physicians* 3, 080–083. <https://doi.org/10.4103/0974-2727.86838>.
- Ragan, M.A., Beiko, R.G., 2009. Lateral genetic transfer: open issues. *Philos Trans R Soc B Biol Sci* 364, 2241–2251. <https://doi.org/10.1098/rstb.2009.0031>.
- Riera Romo, M., Pérez-Martínez, D., Castillo, Ferrer C., 2016. Innate immunity in vertebrates: an overview. *Immunology* <https://doi.org/10.1111/imm.12597>.
- Salcedo, D.E., Lee, J.H., Ha, U.H., Kim, S.P., 2015. The effects of antibiotics on the biofilm formation and antibiotic resistance gene transfer. *Desalin. Water Treat.* 54, 3582–3588. <https://doi.org/10.1080/19443994.2014.932306>.
- Singh, R., Samal, K., Dash, R.R., Bhunia, P., 2019. Vermifiltration as a sustainable natural treatment technology for the treatment and reuse of wastewater: a review. *J. Environ. Manag.* 247, 140–151. <https://doi.org/10.1016/j.jenvman.2019.06.075>.
- Sinha, R.K., Bhamrabe, G., Chaudhari, U., 2008. Sewage treatment by vermifiltration with synchronous treatment of sludge by earthworms: a low-cost sustainable technology over conventional systems with potential for decentralization. *Environmentalist* <https://doi.org/10.1007/s10669-008-9162-8>.
- Sun, M., Chao, H., Zheng, X., Deng, S., Ye, M., Hu, F., 2020. Ecological role of earthworm intestinal bacteria in terrestrial environments: a review. *Sci. Total Environ.* <https://doi.org/10.1016/j.scitotenv.2020.140008>.
- Tehrani, A.H., Gilbride, K.A., 2018. A closer look at the antibiotic-resistant bacterial community found in urban wastewater treatment systems. *Microbiologyopen* <https://doi.org/10.1002/mbo3.589>.
- Tong, J., Tang, A., Wang, H., Liu, X., Huang, Z., Wang, Z., et al., 2019. Microbial community evolution and fate of antibiotic resistance genes along six different full-scale municipal wastewater treatment processes. *Bioresour. Technol.* <https://doi.org/10.1016/j.biortech.2018.10.079>.

- Wang, X., Wang, X., Zhang, Y., Qu, X., Yang, S., 2003. An antimicrobial peptide of the earthworm *Pheretima tschiliensis*: cDNA cloning, expression and immunolocalization. *Biotechnol. Lett.* <https://doi.org/10.1023/A:1024999206117>.
- Wang, L., Zheng, Z., Luo, X., Zhang, J., 2011. Performance and mechanisms of a microbial-earthworm ecofilter for removing organic matter and nitrogen from synthetic domestic wastewater. *J. Hazard. Mater.* <https://doi.org/10.1016/j.jhazmat.2011.08.035>.
- Wang, Y., Xing, M., Yang, J., 2017. Earthworm (*Eisenia fetida*) eco-physiological characteristics in vermifiltration system for wastewater treatment through analyzing differential proteins. *Water Air Soil Pollut.* <https://doi.org/10.1007/s11270-016-3138-y>.
- Wang, J., Chu, L., Wojnárovits, L., Takács, E., 2020. Occurrence and fate of antibiotics, antibiotic resistant genes (ARGs) and antibiotic resistant bacteria (ARB) in municipal wastewater treatment plant: an overview. *Sci. Total Environ.* 744, 140997. <https://doi.org/10.1016/j.scitotenv.2020.140997>.
- Yang, J., Liu, J., Xing, M., Lu, Z., Yan, Q., 2013. Effect of earthworms on the biochemical characterization of biofilms in vermifiltration treatment of excess sludge. *Bioresour. Technol.* <https://doi.org/10.1016/j.biortech.2013.05.099>.